

Case Study: Automated Scoring of Self-Introduction Transcripts Using Rule-Based and NLP Transformers

Introduction & Problem Statement

Analyzing self-introduction transcripts is an important HR exercise. Manual grading cant be scaled properly and also lacks consistency. This case study provides a solution built using Python and Streamlit to automate scoring of transcripts by combining rule-based and transformer-powered NLP features.

Tools and Technology Choices

Streamlit: For faster UI development and visualization.

NLTK: For classic NLP processing (tokenization, sentiment).

PySpellChecker: For error detection in text (spelling/grammar proxy).

sentence-transformers (MiniLM): For semantic similarity via transformer embeddings.

Torch: Backend for running transformer models.

This combination is chosen as they are easy to deploy, compatible with cloud apps, and has capability to demonstrate both traditional and modern NLP techniques.

Solution Architecture

User Flow

1. User pastes their transcript and provides speech timing.

2. The app instantly calculates scores for:

- Content & structure (keywords, salutation, flow)
- Speech rate, clarity, and vocabulary richness
- Grammar/spelling (with PySpellChecker)
- Engagement (NLTK sentiment)
- Semantic similarity: How well the transcript matches rubric content, using transformer embeddings.

Scoring Breakdown

Rule-based: Each rubric condition receives a numeric score. Example: salutation (0-5), keyword coverage (0-30), filler words, vocabulary, etc.

NLP-based: Each rubric statement and the user's transcript are converted to embeddings it determines how "close" in meaning the transcript is to official descriptions. This helps us give a more contextual and flexible grading method.

Implementation

All rules and features were implemented in Python.

python

```
model = SentenceTransformer('all-MiniLM-L6-v2')

def semantic_similarity(text, rubric_list):
    emb = model.encode(text, convert_to_tensor=True)
    rubric_embs = model.encode(rubric_list, convert_to_tensor=True)
    similarities = util.cos_sim(emb, rubric_embs).cpu().numpy().flatten()
    return similarities.max(), rubric_list[similarities.argmax()], similarities
```

Scoring, tokenization, spelling detection, and sentiment analysis were combined in a single Streamlit interface. Requirements are managed with a single requirements.txt file, suitable for cloud deployment.

Global deployment(LINK):-

<https://self-introduction-scorer-qmddynpjiz7yp9audggt8y.streamlit.app/>

Local deployment:-

These 2 lines

```
nltk.download('vader_lexicon', quiet=True, force=True)
```

```
nltk.download('punkt', quiet=True, force=True)
```

can create problem to avoid this we have to do this

```
nltk.download('vader_lexicon')
```

```
nltk.download('punkt')
```

and then we will run this command `streamlit run rubric_scoring.py` and it will open the streamlit interface locally.

Results and Output Samples

Sample outputs included:

Scores for content, keyword presence, flow, grammar, clarity, and engagement. A final semantic similarity score, showing which rubric statement best matches the introduction.

Example:

Highest similarity to rubric: 0.85

Most similar rubric item: "I aspire to be a doctor."

Discussion and Product Thinking

This approach delivers a scalable, flexible, and cloud-compatible way to score personal transcripts.

Advantages include:

- **Automation:** Reduces manual effort and bias.
- **Context Awareness:** Transformer-based similarity adapts to phrasing, not just keyword matching.
- **Extensibility:** New rubric statements and advanced scoring logic can be added easily.
- **Deployment:** Cloud and desktop ready; minimal system requirements.
- **Future scope:** Add audio input and speech-to-text, use classroom dashboards for batch scoring, adapt transformer models for multilingual support.

Conclusion

By combining rule-based logic and transformer NLP models, this project demonstrates effective automation of transcript scoring, moving beyond rigid keyword detection to contextual understanding. This design offers significant value for HR professionals seeking fair, reproducible, and transparent grading.